

SULIT

DP024

Physics 2

Semester 2

Session 2023/2024

2 hours

DP024

Fizik 2

Semester 2

Sesi 2023/2024

2 Jam



KEMENTERIAN PENDIDIKAN

KOLEJ MATRIKULASI KEDAH

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULSI

MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

SET 2

Kertas soalan ini mengandungi 7 halaman bercetak

This question paper consists of 7 printed pages

1 (a) The water waves in Darulaman lake is represented by the equation

$$y = 0.6 \sin (20t - 4x)$$

where x and y are in meters and t is in second.

- (i) Calculate the velocity of the water waves.
- (ii) A surfer is surfing on the water waves in the Darulaman lake. When the surfer is at the position $x = 1.6$ m at the instant time, $t = 60$ s, determine the displacement of the surfer and the speed of the surfer.

[8 marks]

(b) Displacement equation of stationary wave is given by $y = 3 \cos (0.1\pi x) \sin (100\pi t)$

where y and x are in centimetres. Determine

- (i) Amplitude of particle when $x = 2$ cm
- (ii) Wavelength
- (iii) Frequency
- (iv) Speed of wave

[5 marks]

(c)

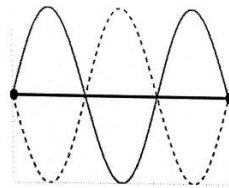


FIGURE 1

Two sinusoidal waves with the same amplitude and wavelength travel in opposite direction along a string that is stretched along an axis. Their resultant wave is shown in **FIGURE 1**. The distance between nodes and adjacent antinodes is 2 cm. The mass of the string is 35 g and the tension force applied is 50 N. Determine the speed of waves travelling along the string

[3 marks]

- 2 (a) A metal wire with a uniform cross-sectional area of 0.08 mm^2 and length of 5 m is applied a potential difference of 8.0 V . Determine resistivity of the wire when the current that flow in the wire is 1.2 A .

[2 marks]

(b)

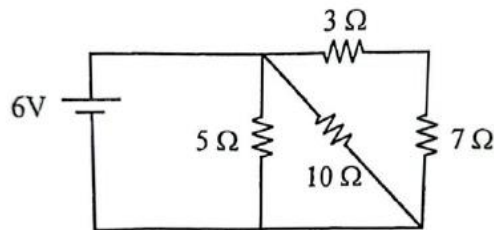


FIGURE 2.1

FIGURE 2.1 shows an arrangement of resistors connected to a battery with emf 6 V . Caculate the

- (i) effecteive resistance.
- (ii) current through the battery.

[6 marks]

(c)

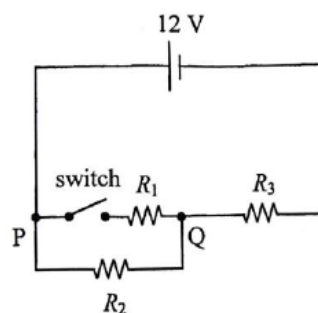


FIGURE 2.2|

FIGURE 2.2 show a circuit consisting of a 12 V battery, a switch and three resistors, $R_1 = 20 \Omega$, $R_2 = 30 \Omega$ and $R_3 = 40 \Omega$. If a voltmeter is connected across PQ and the switch is opened, determine the reading shown on the voltmeter.

[4 marks]

3 a)

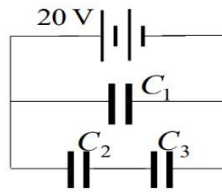


FIGURE 1

FIGURE shows a combination of three capacitors where $C_1 = 100 \mu\text{F}$, $C_2 = 22 \mu\text{F}$ and $C_3 = 47 \mu\text{F}$. A 20 V supply is connected to the combination. Determine

- (i) the effective capacitance in the circuit.
- (ii) the charge stored in the capacitor C_1 .

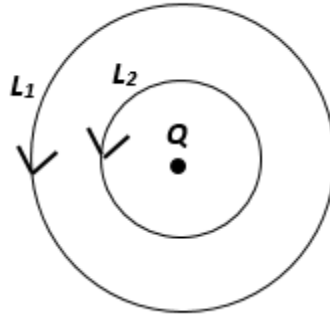
[5 marks]

- b) A $12 \mu\text{F}$ capacitor is charged to a potential difference of 6 V. It is then disconnected and discharges through $20 \text{ k}\Omega$ resistor. Calculate

- (i) the time constant
- (ii) the charge remained in the capacitor after 0.5 s .
- (iii) Sketch a graph charge against time

[5 marks]

4. a) Figure below shows a circular loop L_1 of radius 8 cm carries a 14.0 A current and another loop L_2 of radius 1.8 cm, carries a 4.0 A current is placed at the centre of loop L_1 . What will be the resultant magnetic field at point Q if the two current flow in the same direction?

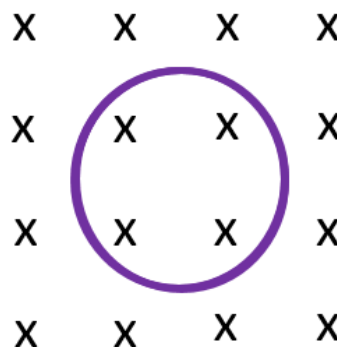


[3 marks]

- b) A 25 turns coils with cross-sectional area of $6.8 \times 10^{-3} \text{ m}^2$ is placed in a uniform magnetic field of 2.5 T. the plane of the coil makes an angle of 40° to the magnetic field. Calculate
- the magnetic flux
 - magnetic flux linkage

[4 marks]

- c) A circular coil of 20 turns and diameter of 5.0 cm is placed perpendicular to a uniform magnetic field of 2.50 T as shown in **FIGURE** below. After 0.30 s, the diameter of the coil is increased to 6.8 cm.
- Calculate the change in the area of the coil.
 - If the coil has a resistance of 1.2Ω , determine the induced current and its direction in the coil.



[5 marks]

- d) A straight conductor of length 30 cm moves in a uniform magnetic field of flux density 40 mT at a constant speed of 20 m s^{-1} . The velocity makes an angle 30° to the field but the conductor is perpendicular to the field. Determine the induced emf experienced by the conductor.

[2 marks]

- 5 a) Siti looks at her face in a make-up mirror. When she is 22.0 cm away from the mirror, her face image is magnified 1.4 times.
- (i) State the type of make-up mirror. Give the reason.
 - (ii) Calculate the radius of curvature of the mirror.
 - (iii) Sketch the ray diagram to show the formation of the image
- [8 marks]
- b) A certain LCD projector contains a single thin lens. An object 24.0 mm high is to be projected so that its real image fills a screen 1.80 m high. The object-to-screen distance is 3.00 m.
- (i) Determine the focal length of the projection lens.
 - (ii) How far from the object should the lens of the projector be placed in order to form the image on the screen?
- [6 marks]

6. (a) A 600 nm sodium light illuminates a double-slit with slit separation 0.5 mm. If the screen is 2.0 m from the slits, calculate the
- (i) distance between central bright to the second order maximum.
 - (ii) distance between central bright to the fifth order minimum.
- [6 marks]
- (b) In a Young's double slit experiment using light of wavelength $6.0 \times 10^{-7} \text{ m}$, the slits were 0.4 mm apart and the distance of the slits to the screen was 1.20 m. Calculate the separation of the fringes.
- [2 marks]
- (c) A 590 nm sodium light illuminates a double-slit with slit separation 0.4 mm. If the screen is 1.5 m from the slits.
- (i) Calculate the distance between two successive dark fringes.
 - (ii) Calculate the separation of the third order minimum from the central bright fringe.
 - (iii) State the effect of the interference pattern when the wavelength of the source is increased.
- [6 marks]